JRC

# JRC ERP — How it works

Jazallah Road Contracting · one system for estimating, site delivery, supply chain, billing, finance, people and plant

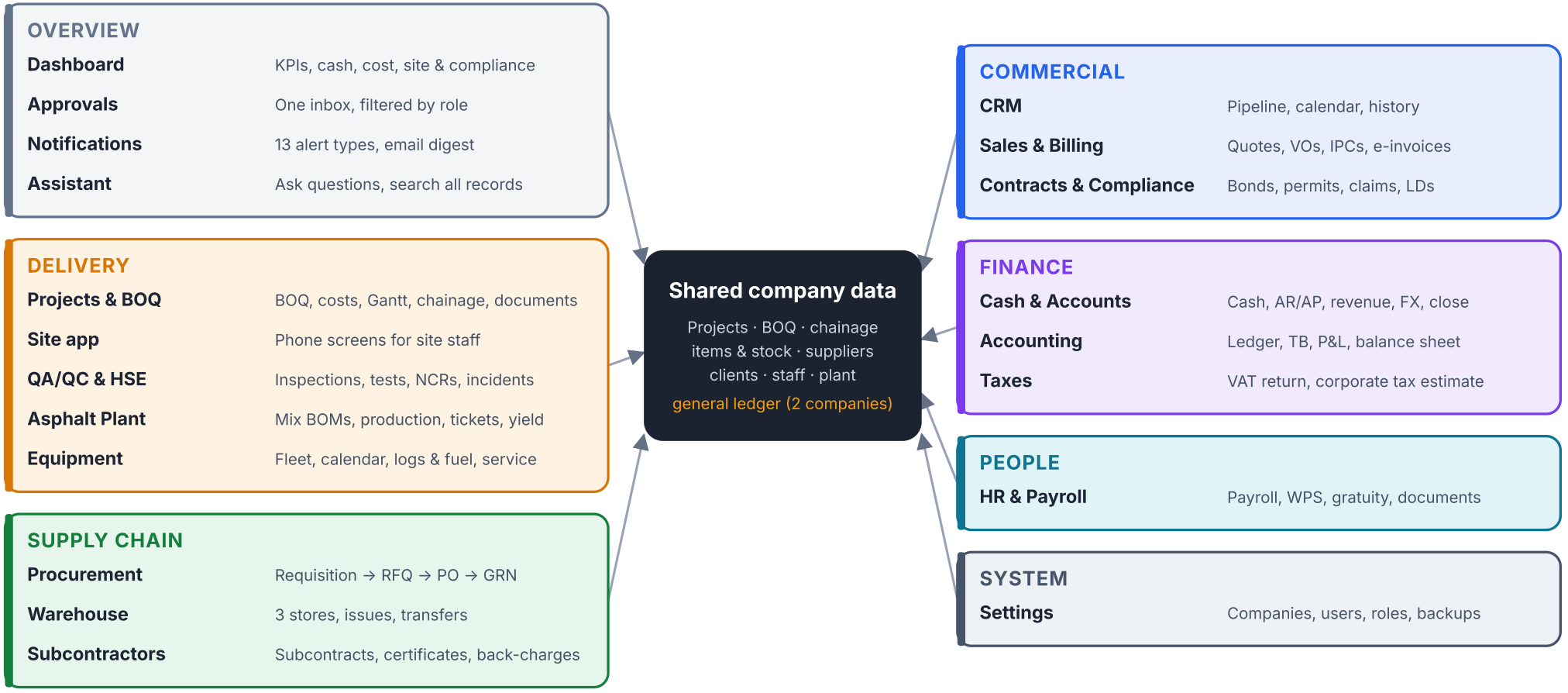

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All company details, people and figures in the system and in this guide are fictional samples. VAT and WPS outputs are illustrations to be checked with your accountant and bank.

# 1 · Module map

20 modules share one set of company data

Every module reads from and writes to the same records: a quantity booked on site, a delivery received in the store, or an hour of overtime feeds straight into costs, invoices, VAT and the dashboard. Nothing is typed twice.

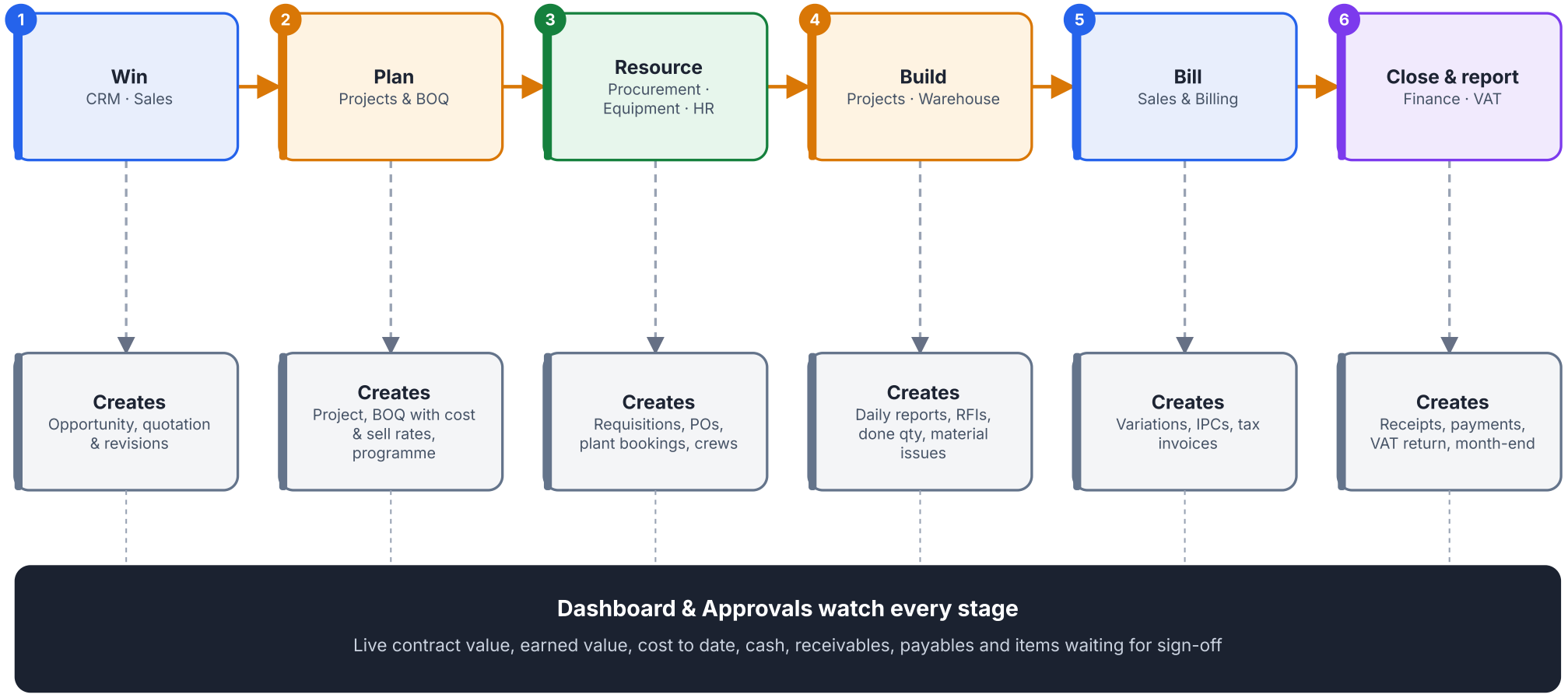



Delivery · Supply chain · Commercial · Finance · People · Overview · System

# 2 · The project lifecycle



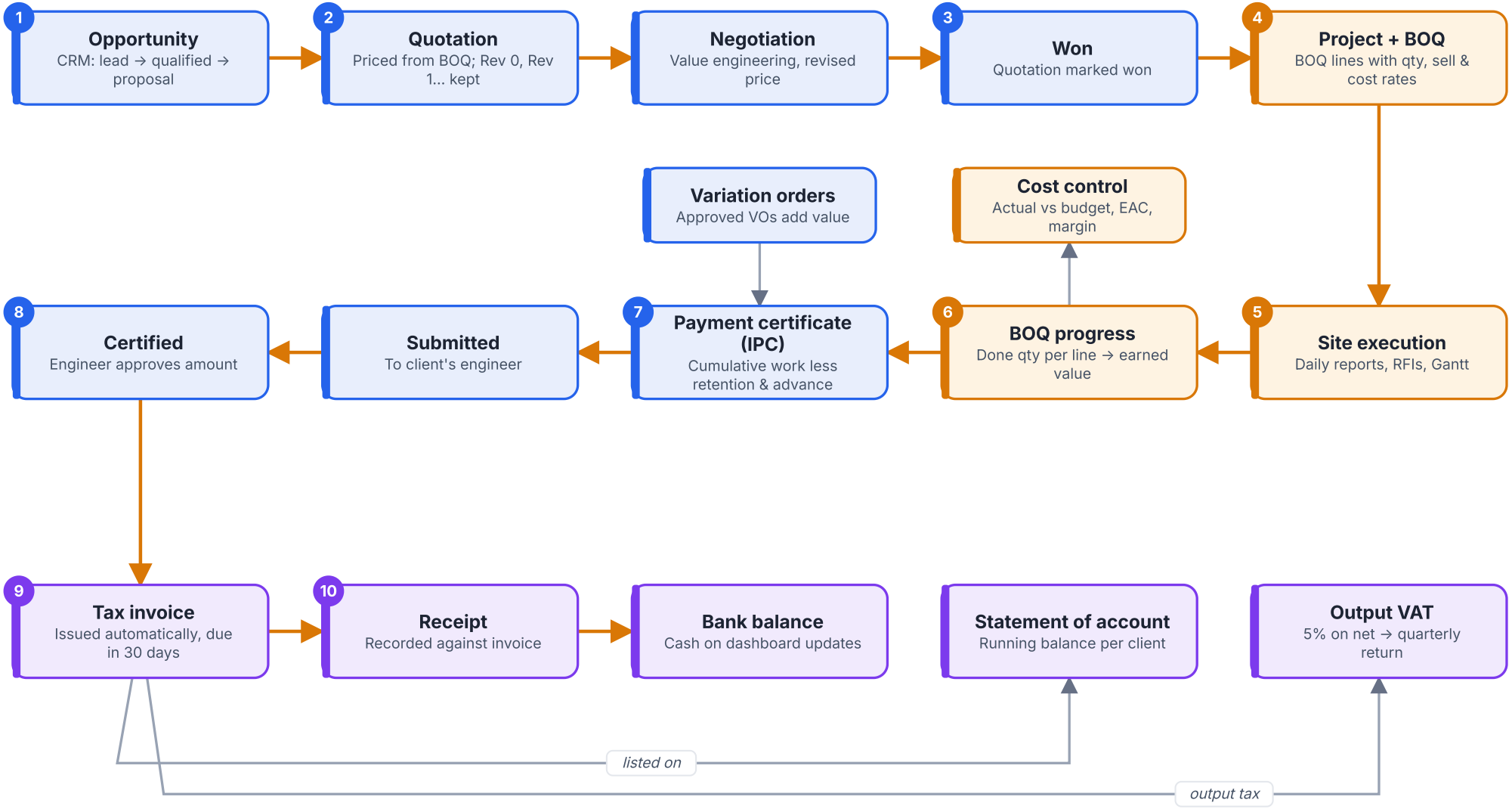
A road project moves through six stages. Each stage is handled by one or more modules; the bottom row shows the records each stage creates.



# 3 · Win work → get paid



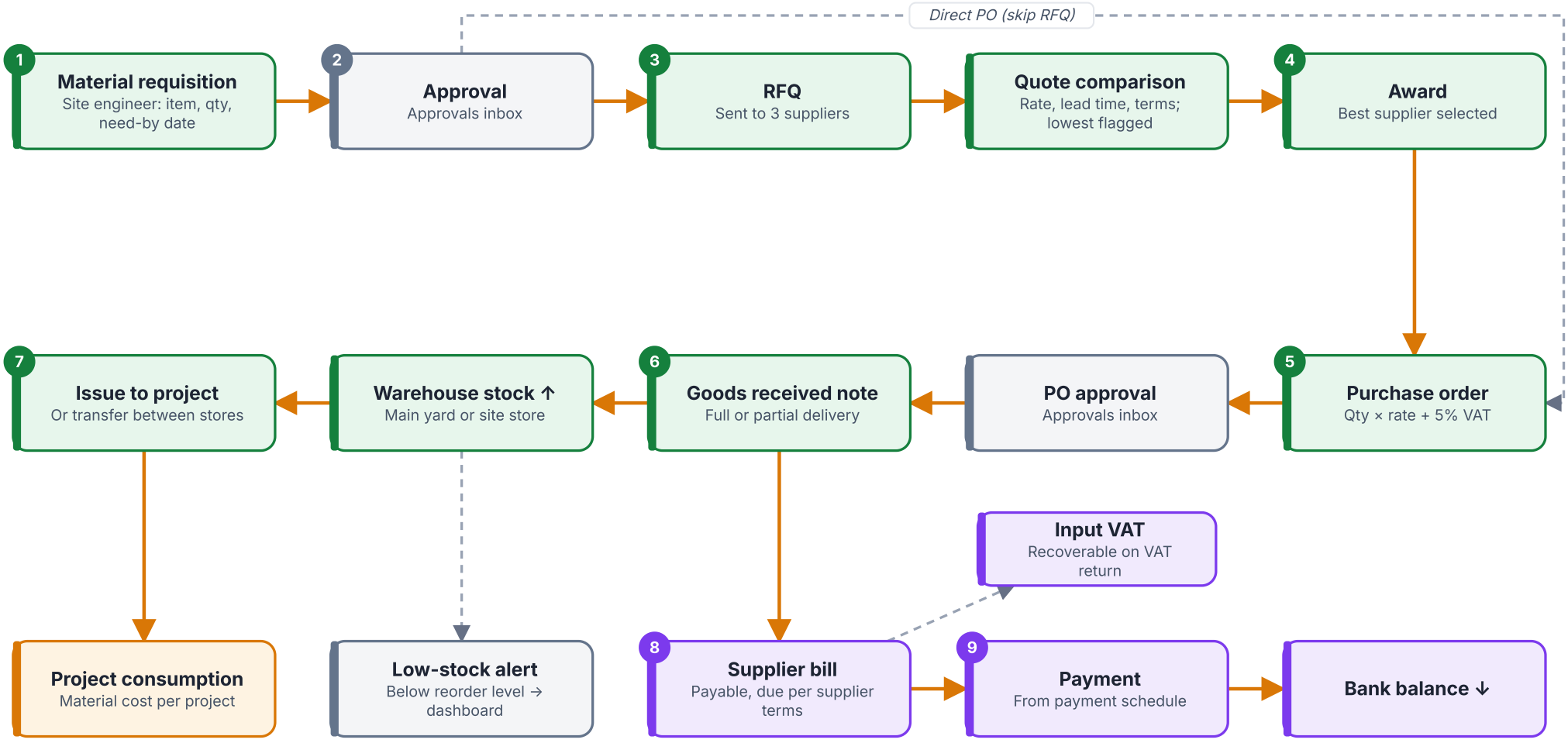
Follow the arrows: an opportunity becomes a priced quotation, then a project with a BOQ. Site progress against the BOQ produces the payment certificate (IPC), which becomes a tax invoice and finally cash in the bank.



# JRC 4 · Buy materials → pay suppliers

Procurement · Warehouse · Payables · VAT

Site asks for material, the office gets competing quotes, the winner gets a PO, the store receives it, and the material is issued to the project where it becomes cost. The supplier's bill flows to payables and its VAT becomes recoverable input tax.



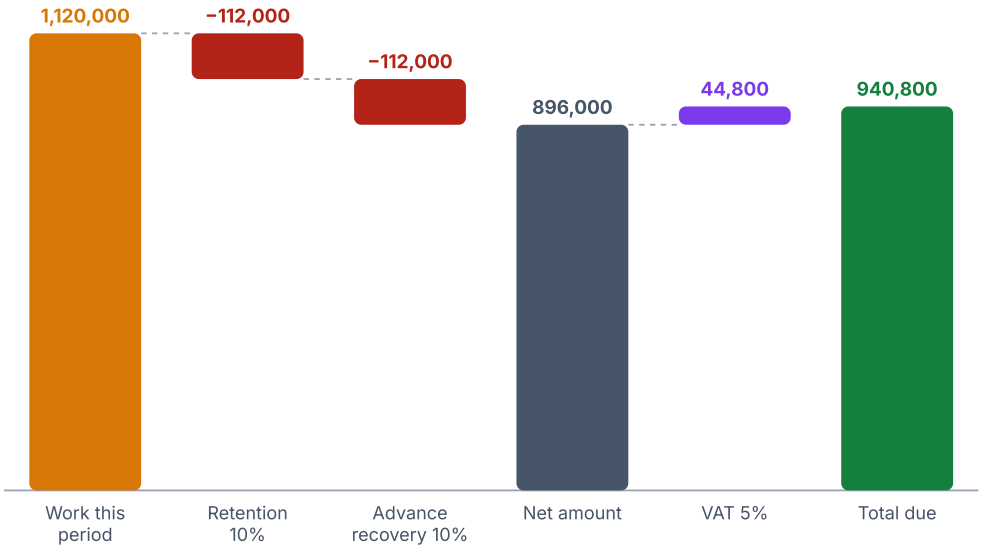
# 5 · Payment certificate (IPC) maths

Worked example: IPC-2401-09, Al Wathba Industrial Access Road

**Cumulative work to date: AED 8,040,000**

Certified before (IPC-08): 6,920,000 | 1,120,000 this period

**This certificate (AED)**




Retention is held by the client (tracked under Receivables) and released at handover / end of defects liability.

## Where cumulative work comes from

```
Earned value = Σ (BOQ done qty × sell rate)
Cumulative work = Earned value + approved VOs × % complete
```

- Site staff update **done quantity** per BOQ line.
- Approved **variation orders** are added pro-rata to progress.
- "New IPC from progress" takes this figure automatically.

## Deductions (set in Settings)

```
Retention = 10% × work this period
Advance recovery = 10% × work this period
Net = work − retention − advance recovery
VAT = 5% × net · Total due = net + VAT
```

- Retention is tracked as money held by the client and released at handover / end of defects liability.

## Status path

- **Draft** → **Submitted** to the engineer → **Certified** (a tax invoice is issued automatically) → **Paid** when the receipt clears the invoice.

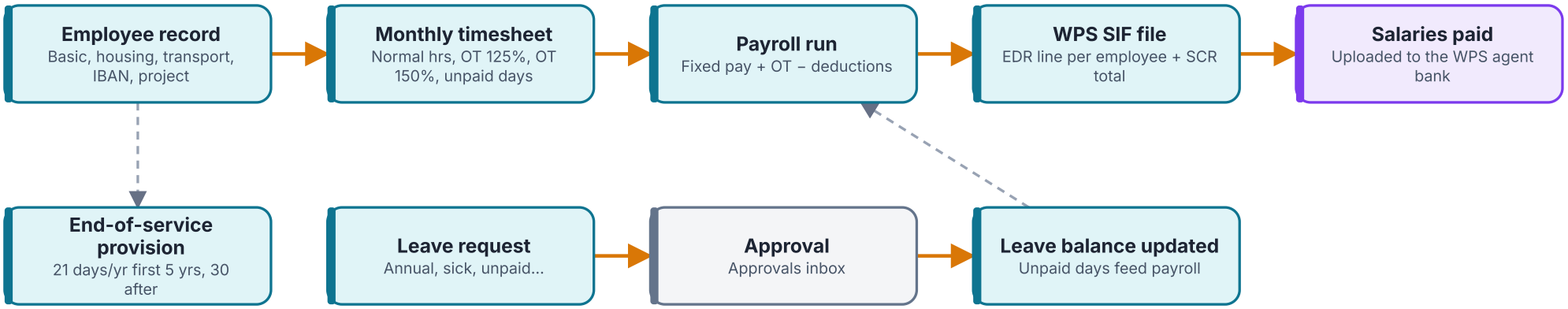
## Revenue recognition (Finance)

```
% complete = cost to date ÷ estimated total cost
Revenue to date = revised contract × % complete
Contract asset / (liability) = revenue − certified
```

# 6 · People & plant

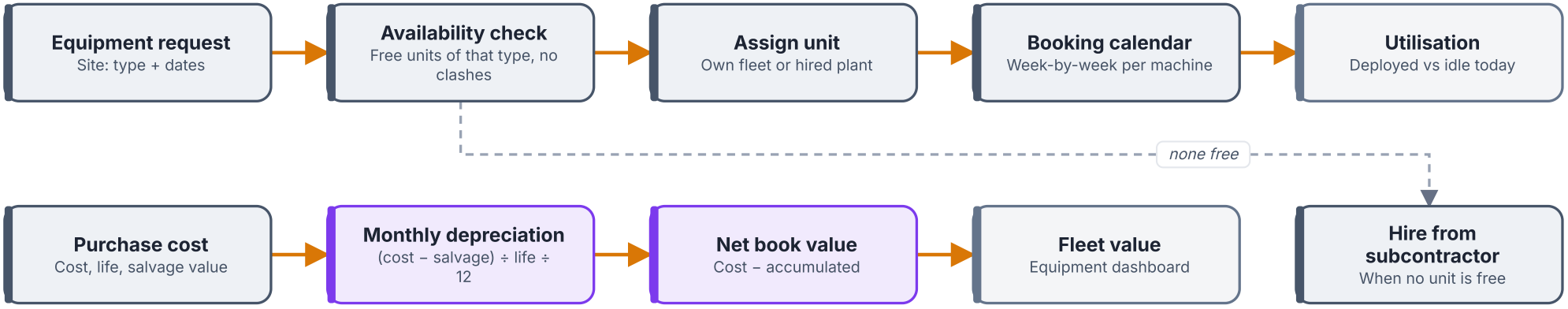


Hours worked become payroll and a bank-ready WPS file; service years drive the end-of-service provision. Equipment is requested by site, checked for clashes, booked on the calendar and depreciated monthly.




HR & PAYROLL ↑

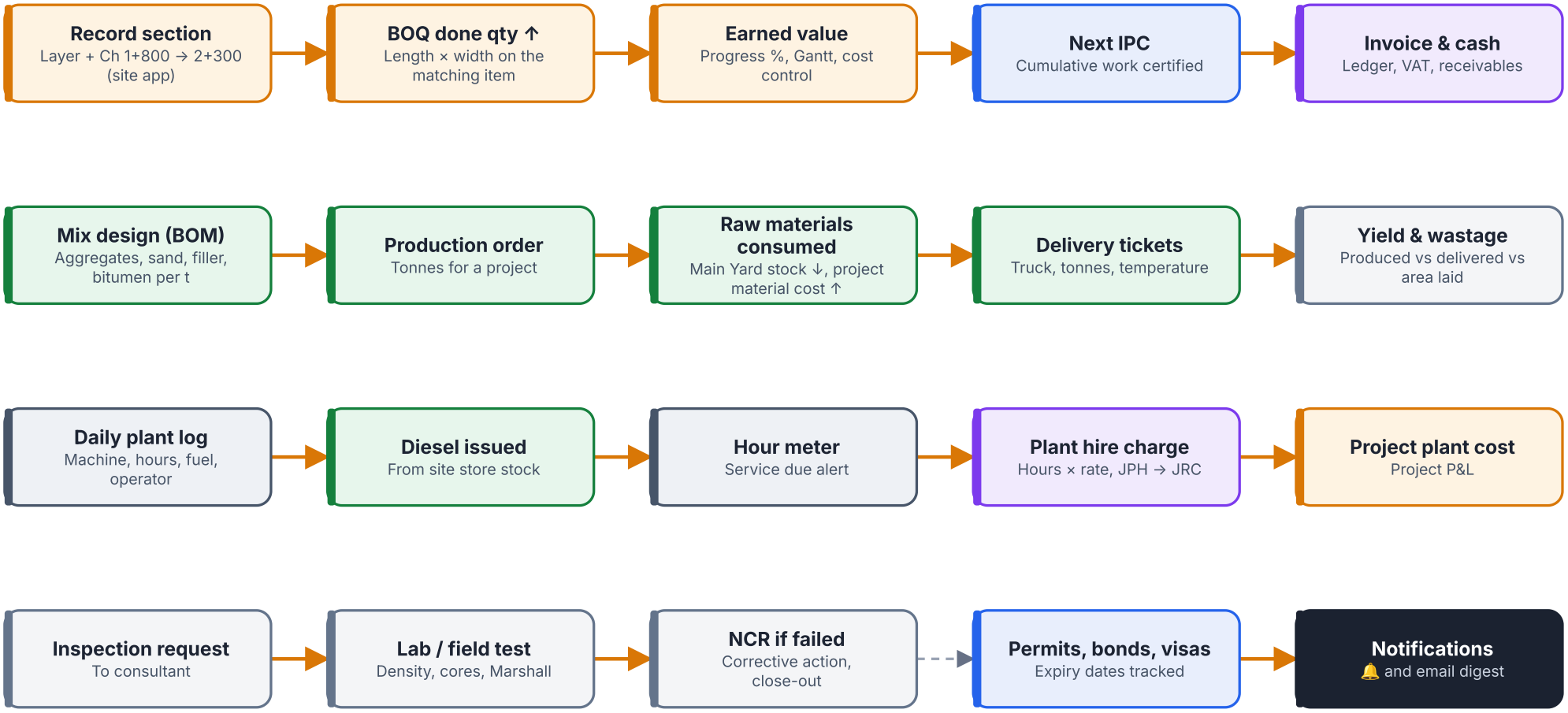
EQUIPMENT ↓



# 7 · Road operations

Chainage · Asphalt plant · Plant logs · QA/QC · Compliance

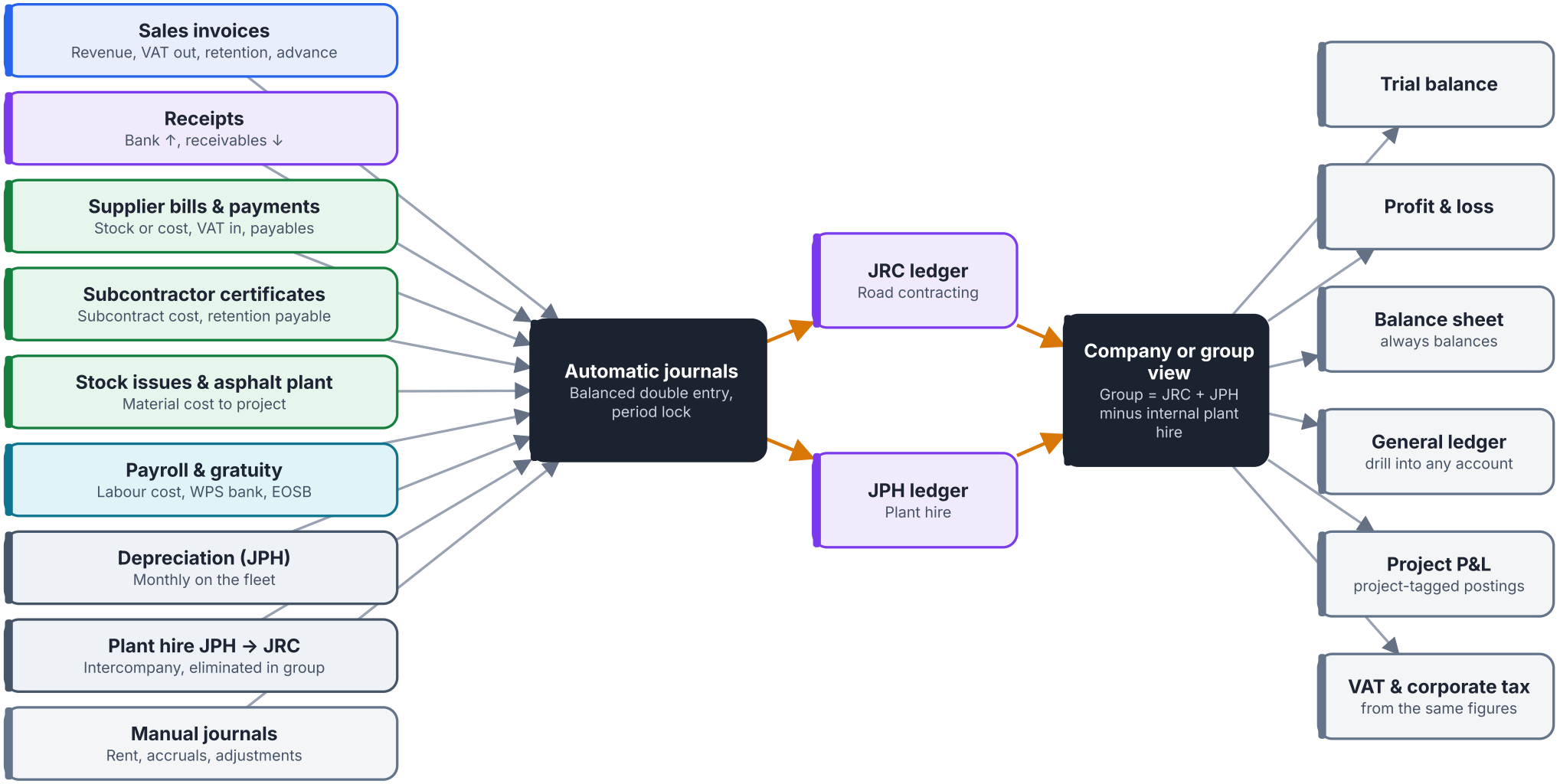
The road-specific modules feed the same records as everything else: a section recorded by chainage raises BOQ progress and the next payment certificate; asphalt production consumes stock; plant hours become project cost; failed tests and expiring permits raise alerts.



# 8 · Accounting & group companies

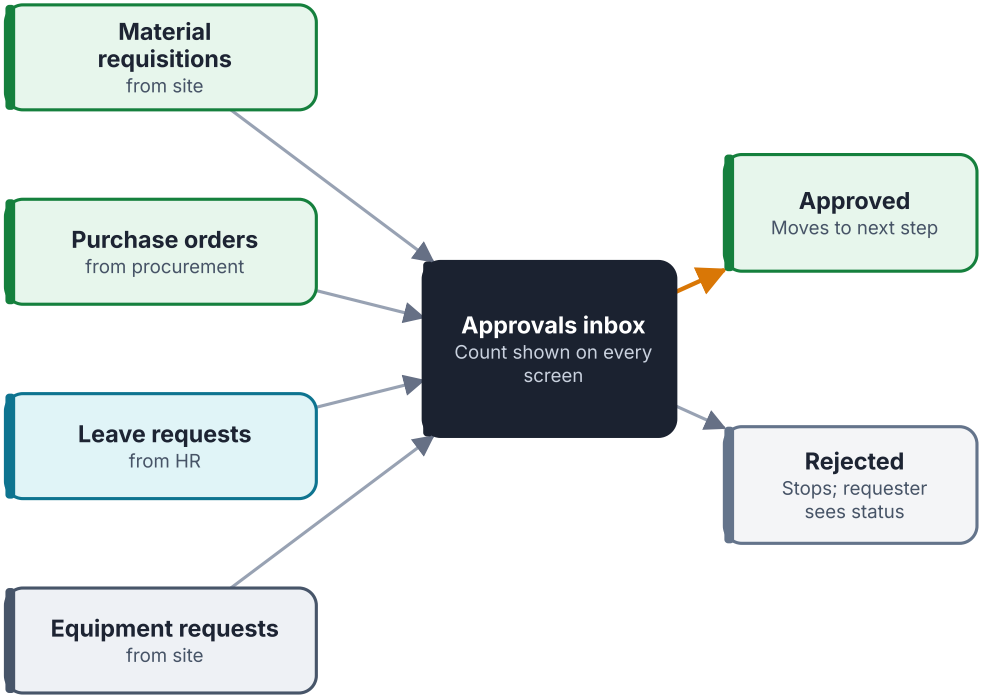
Every transaction posts itself to the ledger

Nobody keys journals for day-to-day work: invoices, receipts, bills, payroll, stock issues, depreciation and plant hire each create a balanced double entry. Two companies keep separate books — Jazallah Road Contracting and Jazallah Plant Hire (which owns the fleet and hires it to JRC) — and the group view removes the internal hire.



# 9 · Approvals, alerts & what updates what






Approver per document and amount is set in Settings → Approval workflow

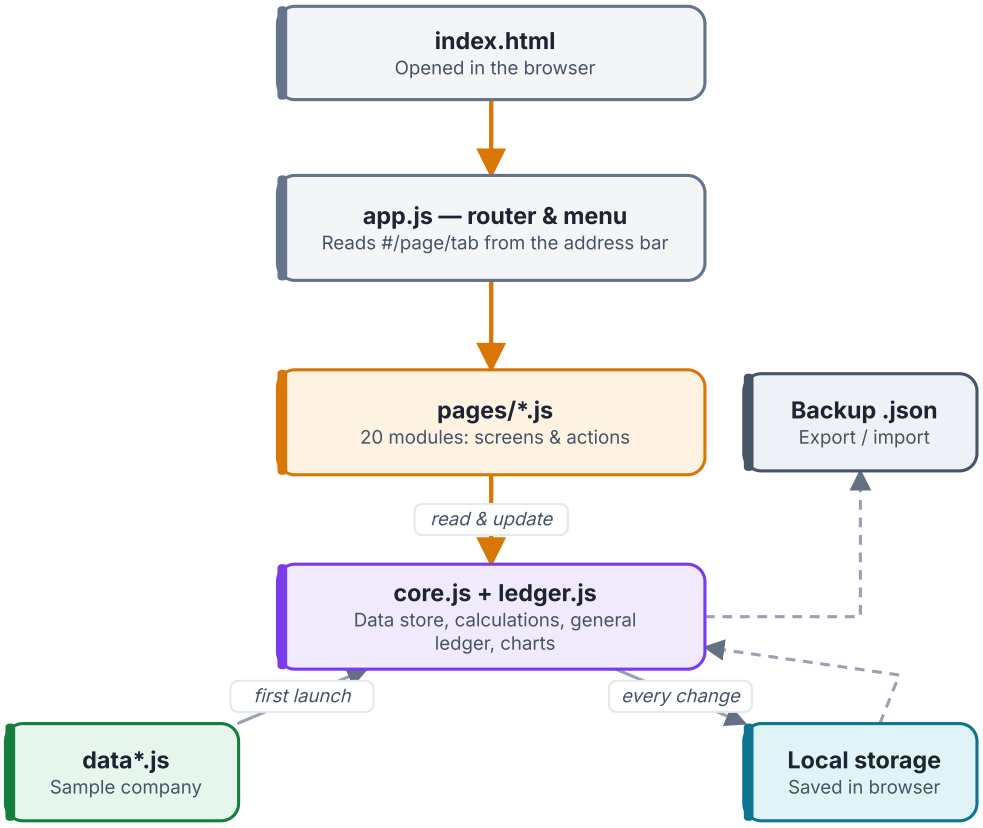
| When you... | The ERP automatically... |
|---|---|
| Update BOQ done quantity | Recalculates earned value, progress %, Gantt bars, cost variance, forecast margin and the next IPC amount |
| Award an RFQ | Creates a PO (pending approval); adds a new supplier to the register if needed |
| Post a goods received note | Adds stock to the chosen warehouse, marks PO received, creates a supplier bill in payables |
| Issue stock to a project | Reduces warehouse stock, adds to project consumption, triggers low-stock alert if below reorder level |
| Certify an IPC | Issues a tax invoice due in 30 days; it appears in receivables, statement of account and output VAT |
| Record a receipt | Increases the bank balance, reduces receivables, marks the IPC paid when fully settled |
| Pay a supplier bill | Reduces the bank balance and clears the payable |
| Edit timesheet hours | Recalculates overtime pay, payroll total and the WPS file |
| Approve annual leave | Deducts days from the employee's leave balance |
| Assign equipment | Books it on the calendar (blocking clashes) and updates fleet utilisation |
| Approve a variation order | Raises the revised contract value and future IPC amounts |
| Record a chainage section | Adds length × width to the matching BOQ line, moves the schedule bar and earned value |
| Produce an asphalt order | Consumes aggregates, sand, filler and bitumen from Main Yard; posts material cost to the project |
| Log plant hours with fuel | Issues diesel from the site store, advances the hour meter (service alerts), charges plant hire to the project |
| Approve a subcontractor certificate | Creates a payable net of retention and back-charges; posts subcontract cost |
| Reject an inspection request | Optionally raises an NCR that stays open until closed out |

### Notifications (🔔)

- Overdue invoices and bills, low stock, items to approve
- Bonds, permits and staff documents nearing expiry
- Failed tests, open NCRs, plant service due, delay (LD) exposure, milestones due

# 10 · Under the hood






## How it runs

- Plain HTML, CSS and JavaScript — no installation, no server, no build step.
- Open **erp/index.html** in any modern browser.
- Each screen is a page module; the address bar hash (e.g. **#/projects/P-2401/chainage**) decides which one shows.
- The Site app can be installed on a phone's home screen when the ERP is served over the web.

## Where the data lives

- First launch loads the sample company from **data.js**.
- Every change is saved in the browser's local storage, so it survives a refresh.
- Settings → **Export backup** downloads everything as one JSON file; **Import** restores it; **Reset** returns to the sample.

## Limits of the demo

- Users are switched from a menu, not real logins; roles limit what each person sees but are not enforced by a server.
- E-invoices are generated as XML files; email opens your mail app.
- Data is per browser; colleagues don't see each other's changes.

## Path to production

- Move the data store to a hosted database (e.g. Supabase) with user accounts and role-based access.
- Keep the same screens and calculations — only the store layer changes.
- Add email notifications, a mobile layout for site staff and document attachments.